Mathematics - K-12 Level Assessment 12 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. The practice range at a golf course charges \$4.00 for a bucket of 40 golf balls. At this rate, how much will a bucket of 100 golf balls cost?
   (a) \$10.00
   (b) \$10.50
   (c) \$13.50
   (d) \$16.00

2. Find the median in the set of data 23, 13, 18, 29, 32, 25.
   (a) 18
   (b) 24
   (c) 25
   (d) 29

3. Which expression is equivalent to $5(4\ x + 3) - 2\ x$?
   (a) $18\ x{+}15$
   (b) $18\ x{+}3$
   (c) $7\ x{+}8$
   (d) $2\ x{+}8$

4. What is the value of the expression $2(3(4^2 + 1)) - 2^3$?
   (a) 156
   (b) 110
   (c) 94
   (d) 48

5. Which group of numbers is in order from least to greatest?
   (a) 0.25 1.6 1.0
   (b) 1.0 0.25 1.6
   (c) 0.25 1.0 1.6
   (d) 1.6 1.0 0.25

True / False (10 marks)

Answer True or False

6. True or False: A correlation of 0.6 explains six times the percentage of variation in y compared to a correlation of 0.3.
7. True or False: Tara will take 1 hour and 12 minutes to develop 8 rolls of film at her current rate.
8. True or False: It takes about 7.69 years for two thirds of a radioactive substance with an 8-year half-life to decay.
9. True or False: Marcus took 16 minutes to wash his car, using his quarters to pay for the time spent.
10. True or False: Joe finished the race at 3:58 p.m., 4 minutes before Mike, who finished at 4:02 p.m.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. We define two sequences of vectors $(\mathbf{v}_n)$ and $(\mathbf{w}_n)$ as follows: First, $\mathbf{v}_0 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$, $\mathbf{w}_0 = \begin{pmatrix} 4 \\ 0 \end{pmatrix}$. Then for all $n \geq 1$, $\mathbf{v}_n$ is the projection of $\mathbf{w}_{n-1}$ onto $\mathbf{v}_0$, and $\mathbf{w}_n$ is the projection of $\mathbf{v}_n$ onto $\mathbf{w}_0$. Find
$$\mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3 + \cdots .$$
12. Calculate the total cost of purchasing 4 magazines at 2.99$each and 3 books at$6.99 each. Explain your calculation process and discuss how you arrived at the final amount.

End of Paper